



FABLESS SEMICONDUCTOR ASSOCIATION

TSMC PDK Checklist

Foundry - TSMC
 Process – 0.18um MIXED SIGNAL GENERAL PURPOSE
 II 1P6M/1P5M/1P4M SALICIDE 1.8V/3.3V PDK
 PDK Revision – Version 1.2A, 04/22/2014
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PDK Support Contact

E-mail pdk@tsmc.com

Foundry Process Documents

Document	Document Number & Title	Section	Revision	Date
Design Manual(Devices)	T-018-CM-SP-007		1.2	2010/06/21
Design Layout Rules	T-018-LO-DR-001		2.10	2012/11/08
Spice Model	T-018-CM-SP-007		1.2	2010/06/21
DRC	T-018-LO-DR-001-U1		2.10a	2013/01/24
	T-018-LO-DR-001-C1		2.10a	2012/11/14
LVS	T-018-CM-SP-007-U1		1.2a	2013/01/28
	T-018-CM-SP-007-C1		1.2a	2013/01/28
Parasitic Extraction	T-018-CM-SP-007-V1		1.2b	2014/02/13
	T-018-CM-SP-007-X1		1.2b	2013/06/17
Layer Map	T-018-LO-LE-004		2.10a	2013/02/25



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EDA Tools Supported and Verified for Use with this PDK

Type	Vendor and Tool	Version	Version Date
Schematic	Cadence Design Systems, Inc / Composer	IC6.1.5.500.13	
Simulation Control	Cadence Design Systems, Inc / Analog Design Environment	IC6.1.5.500.13	
Layout Editor	Cadence Design Systems, Inc / Virtuoso VirtuosoXL	IC6.1.5.500.13	
Circuit Simulator (A)	Cadence Design Systems, Inc / Spectre	7.2.0.477.isr16	
Circuit Simulator (B)	Synopsys / Hspice – HspiceD	G-2012.06	
Circuit Simulator (C)	Cadence Design Systems, Inc / Verilog	08.20.027	
Circuit Simulator (D)	Mentor Graphics Corporation, Inc / Eldo	2010.1	
DRC Checker	Cadence Design Systems, Inc / Assura Mentor Graphics Corporation, Inc / Calibre	4.1_USR2_HF9 2010.4_26.16	
LVS Checker	Cadence Design Systems, Inc / Assura Mentor Graphics Corporation, Inc / Calibre	4.1_USR2_HF9 2010.4_26.16	
Parasitic Extractor	Cadence Design Systems, Inc / Assura Mentor Graphics Corporation, Inc / Calibre	4.1_USR2_HF9 2010.4_26.16	



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MIXED SIGNAL PDK Checklist

Foundry - TSMC

Process – 0.18um MIXED SIGNAL GENERAL PURPOSE II

1P6M/1P5M/1P4M SALICIDE 1.8V/3.3V PDK

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Device Type	Device Name	Comment	Terminals	Symbol	Spice-Mod	1/f Noise	HF Noise	Stat Mod	Sim-Net-A	Sim-Net-B	Sim-Net-C	Sim-Net-D	LVS Net	SDL Net	GDS	P-Params	Sim-Test-A	Sim-Test-B	Sim-Test-C	Sim-Test-D	DRC Test	LVS Test	Pcell Test
MOS	nmos2v		4	X	X				X	X	X	X	X		X	46	X	X	X	X	X	X	X
	nmos2v_mac		4	X	X			X	X	X	X	X	X		X	46	X	X	X	X	X	X	X
	nmos2vdnw		4	X	X				X	X	X	X	X		X	46	X	X	X	X	X	X	X
	nmos3v		4	X	X				X	X	X	X	X		X	46	X	X	X	X	X	X	X
	nmos3v_mac		4	X	X			X	X	X	X	X	X		X	46	X	X	X	X	X	X	X
	nmos3vdnw		4	X	X				X	X	X	X	X		X	46	X	X	X	X	X	X	X
	nmosmvt2v		4	X	X				X	X	X	X	X		X	46	X	X	X	X	X	X	X
	nmosmvt2v_mac		4	X	X			X	X	X	X	X	X		X	46	X	X	X	X	X	X	X
	nmosnvt2v		4	X	X				X	X	X	X	X		X	46	X	X	X	X	X	X	X
	nmosnvt2v_mac		4	X	X			X	X	X	X	X	X		X	46	X	X	X	X	X	X	X
	nmosnvt3v		4	X	X				X	X	X	X	X		X	46	X	X	X	X	X	X	X
	nmosnvt3v_mac		4	X	X			X	X	X	X	X	X		X	46	X	X	X	X	X	X	X
	nmosmvt3v		4	X	X				X	X	X	X	X		X	46	X	X	X	X	X	X	X
	nmosmvt3v_mac		4	X	X			X	X	X	X	X	X		X	46	X	X	X	X	X	X	X
	pmos2v		4	X	X				X	X	X	X	X		X	46	X	X	X	X	X	X	X
	pmos2v_mac		4	X	X			X	X	X	X	X	X		X	46	X	X	X	X	X	X	X
	pmos3v		4	X	X				X	X	X	X	X		X	46	X	X	X	X	X	X	X
	pmos3v_mac		4	X	X			X	X	X	X	X	X		X	46	X	X	X	X	X	X	X
	pmosmvt2v		4	X	X				X	X	X	X	X		X	46	X	X	X	X	X	X	X
	pmosmvt2v_mac		4	X	X			X	X	X	X	X	X		X	46	X	X	X	X	X	X	X
DIO	dioden		2	X	X				X	X	X	X	X		X	3	X	X	X	X	X	X	X
	dioden3v		2	X	X				X	X	X	X	X		X	3	X	X	X	X	X	X	X
	diodenw		2	X	X				X	X	X	X	X		X	3	X	X	X	X	X	X	X
	diodenw3v		2	X	X				X	X	X	X	X		X	3	X	X	X	X	X	X	X
	diodep		2	X	X				X	X	X	X	X		X	3	X	X	X	X	X	X	X
	diodep3v		2	X	X				X	X	X	X	X		X	3	X	X	X	X	X	X	X
	ndio_m		2	X	X				X	X	X	X	X		X	3	X	X	X	X	X	X	X
	pdio_m		2	X	X				X	X	X	X	X		X	3	X	X	X	X	X	X	X
	pdio_3m		2	X	X				X	X	X	X	X		X	3	X	X	X	X	X	X	X
CAP	crtmom		3	X	X			X	X	X	X	X	X		X	9	X	X	X	X	X	X	X
	mimcap_1p0_sin		2	X	X			X	X	X	X	X	X		X	8	X	X	X	X	X	X	X
	mimcap_1p0_sin_3t		3	X	X			X	X	X	X	X	X		X	8	X	X	X	X	X	X	X
	mimcap_2p0_sin		2	X	X			X	X	X	X	X	X		X	8	X	X	X	X	X	X	X
	mimcap_2p0_sin_3t		3	X	X			X	X	X	X	X	X		X	8	X	X	X	X	X	X	X

	cfmom		3	X	X			X	X	X	X	X	X		X	8	X	X	X	X	X	X	X
	cfmom_mx		5	X	X			X	X	X	X	X	X		X	8	X	X	X	X	X	X	X
RES	rm1		2	X	X				X	X	X	X	X		X	5	X	X	X	X	X	X	X
	rm2		2	X	X				X	X	X	X	X		X	5	X	X	X	X	X	X	X
	rm3		2	X	X				X	X	X	X	X		X	5	X	X	X	X	X	X	X
	rm4		2	X	X				X	X	X	X	X		X	5	X	X	X	X	X	X	X
	rm5		2	X	X				X	X	X	X	X		X	5	X	X	X	X	X	X	X
	rmt		2	X	X				X	X	X	X	X		X	5	X	X	X	X	X	X	X
	rnhpoly		2	X	X			X	X	X	X	X	X		X	9	X	X	X	X	X	X	X
	rnhpoly_dis		3	X	X			X	X	X	X	X	X		X	9	X	X	X	X	X	X	X
	rnplus		3	X	X			X	X	X	X	X	X		X	9	X	X	X	X	X	X	X
	rnplus_2t		2	X	X			X	X	X	X	X	X		X	9	X	X	X	X	X	X	X
	rnlpoly		2	X	X			X	X	X	X	X	X		X	9	X	X	X	X	X	X	X
	rnlpoly_dis		3	X	X			X	X	X	X	X	X		X	9	X	X	X	X	X	X	X
	rnplus		3	X	X			X	X	X	X	X	X		X	9	X	X	X	X	X	X	X
	rnplus_2t		2	X	X			X	X	X	X	X	X		X	9	X	X	X	X	X	X	X
	rnwell		2	X	X				X	X	X	X	X		X	9	X	X	X	X	X	X	X
	rnwod		3	X	X				X	X	X	X	X		X	9	X	X	X	X	X	X	X
	rnwod_2t		2	X	X				X	X	X	X	X		X	9	X	X	X	X	X	X	X
	rnwsti_m		3	X	X				X	X	X	X	X		X	9	X	X	X	X	X	X	X
	rphpoly		2	X	X			X	X	X	X	X	X		X	9	X	X	X	X	X	X	X
	rphpoly_dis		3	X	X			X	X	X	X	X	X		X	9	X	X	X	X	X	X	X
	rphripoly		2	X	X			X	X	X	X	X	X		X	9	X	X	X	X	X	X	X
	rphripoly_dis		3	X	X			X	X	X	X	X	X		X	9	X	X	X	X	X	X	X
	rpplus		3	X	X			X	X	X	X	X	X		X	9	X	X	X	X	X	X	X
	rpplus_2t		2	X	X			X	X	X	X	X	X		X	9	X	X	X	X	X	X	X
	rpplus		3	X	X			X	X	X	X	X	X		X	9	X	X	X	X	X	X	X
	rpplus_2t		2	X	X			X	X	X	X	X	X		X	9	X	X	X	X	X	X	X
VAR	mos_var_b		2	X	X				X	X	X	X	X		X	14	X	X	X	X	X	X	X
	mos_var_b3		2	X	X				X	X	X	X	X		X	14	X	X	X	X	X	X	X
BJT	npn	1	3	X	X				X	X	X	X	X		X	3	X	X	X	X	X	X	X
	npn_mis	1	3	X	X			X	X	X	X	X	X		X	3	X	X	X	X	X	X	X
	vpnp	1	3	X	X				X	X	X	X	X		X	3	X	X	X	X	X	X	X
	vpnp3	1	3	X	X				X	X	X	X	X		X	3	X	X	X	X	X	X	X
	vpnp_mis	1	3	X	X			X	X	X	X	X	X		X	3	X	X	X	X	X	X	X
SPE	dio_dnwpsub	2	2	X					X	X	X	X	X							X			
	dio_pwdnw	2	2	X					X	X	X	X	X							X			
	diodesd3v	2	2	X					X	X	X	X	X							X			

Comments

1. The npn/npn_mis /vpnp/vpnp3 /vpnp_mis cells don't have the layout view. It will be automatically generated during the schematic driven layout procedures.
2. This PDK only provides front-end information for these devices. Users have to provide the layouts and set those parameters manually depending on the layouts wanted.